

42.510 Statistical Learning for Data Science

Unsupervised learning – clustering, recommendation systems

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Week 8, Lecture 2

Outline

- Recommendation systems
- Collaborative filtering
- R implementation

Recommendation systems

- Personalize the user experience for online applications.
- Leverage data on items and customers (e.g., likes, purchase history).
- A key challenge: they must be fast and accurate.
- Common underlying analytics: clustering, collaborative and content filtering.

Recommendation systems

Collaborative filtering: Recommendations are based on attributes of users. Each user is represented by a vector of items where the i -th entry gives the customer's rating of the i -th item. This vector will typically have many empty entries (only a small fraction of the items is ranked or purchased).

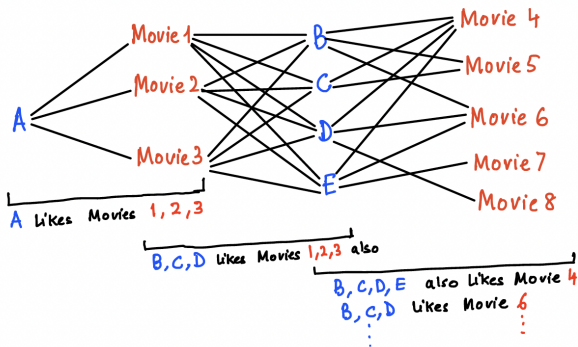
Recommendation systems

Content filtering: Recommendations are made based on attributes of items. Each item is represented by a set of attributes (e.g., genre of movie, keywords, or webpage).

Hybrid: This is a combination of both collaborative and content filtering.

Collaborative filtering

Intuition



Collaborative filtering

Baseline model

Predict the rating based on the item's average popularity:

$$b_{ui} = \bar{r}_i$$

Collaborative filtering

User-based collaborative filtering

Based on:

- 1 Identifying users whose ratings are similar to those of the active user, and,
- 2 Using their ratings on other items to predict what the active (current) user will like.

Collaborative filtering

User-based collaborative filtering

To measure the similarity between users u and v , we can use the *Pearson correlation coefficient*

$$S_{uv} = \frac{\sum_{i \in I_u \cap I_v} (r_{ui} - \bar{r}_u)(r_{vi} - \bar{r}_v)}{\sqrt{\sum_{i \in I_u \cap I_v} (r_{ui} - \bar{r}_u)^2 \sum_{i \in I_u \cap I_v} (r_{vi} - \bar{r}_v)^2}}.$$

Collaborative filtering

User-based collaborative filtering

To predict the rating, we can use the information on a set of neighbors of user u :

$$p_{ui} = \frac{\sum_{v \in N_u} r_{vi}}{|N_u|}.$$

Collaborative filtering

User-based collaborative filtering

A slightly more sophisticated model:

$$p_{ui} = \frac{\sum_{v \in N_u} S_{uv} r_{vi}}{\sum_{v \in N_u} S_{uv}}.$$

Collaborative filtering

User-based collaborative filtering

A modelling trick, normalization:

$$p_{ui} = \bar{r}_u + \frac{\sum_{v \in N_u} S_{uv}(r_{vi} - \bar{r}_v)}{\sum_{v \in N_u} S_{uv}}.$$

Collaborative filtering

Assessing performance

$$RMSE_u = \sqrt{\sum_i \frac{(p_{ui} - r_{ui})^2}{n}}.$$

R implementation

`ratings` **dataset**

Data preparation

R implementation

1. Baseline model 1

2. Baseline model 2

3. User-based model

3. User-based model (cont'd)

References

- Teaching notes